

CHIANG RAI, Thailand

WMO: 483030

Lat: 19.9614N

Lon: 99.8811E

Elev: 393

StdP: 96.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.2	(c) 12.7	(d) 7.6	(e) 6.8	(f) 21.7	(g) 9.3	(h) 7.6	(i) 19.9	(j) 4.7	(k) 20.9	(l) 4.1	(m) 25.0	(n) 0.1	(o) 20	(p) 0.266

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 7.1	(c) 36.4	(d) 23.4	(e) 35.2	(f) 23.4	(g) 34.0	(h) 23.6	(i) 26.5	(j) 31.6	(k) 26.1	(l) 31.2	(m) 25.8	(n) 30.7	(o) 1.9	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 25.2	(b) 21.3	(c) 28.6	(d) 24.9	(e) 20.9	(f) 28.3	(g) 24.6	(h) 20.6	(i) 28.0	(j) 85.4	(k) 31.8	(l) 84.0	(m) 31.4	(n) 82.7	(o) 30.9	(p) 29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.4	3.8	3.3	9.0	38.1	1.4	1.2	8.0	39.0	7.1	39.7	6.3	40.4	5.3	41.2
			8.4	27.5	1.4	0.7	7.4	28.0	6.6	28.4	5.8	28.8	4.8	29.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.7	21.4	23.1	25.7	28.0	28.2	28.1	27.5	27.3	27.2	26.4	24.0	21.6
	DBStd	3.00	1.85	1.69	1.86	1.89	1.95	1.25	1.43	1.20	1.24	1.43	2.10	2.46
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	8	3	0	0	0	0	0	0	0	0	0	0	5
	CDD10.0	5734	352	368	486	541	564	543	542	536	517	507	420	358
	CDD18.3	2701	97	134	228	291	306	293	283	277	267	249	171	104
	CDH23.3	24628	708	1215	2226	3133	3158	2979	2642	2466	2336	1934	1169	662
	CDH26.7	9532	179	501	1115	1647	1424	1099	841	728	779	673	384	161
Wind	WSAvg	0.7	0.5	0.6	0.7	0.8	0.9	0.9	0.9	0.8	0.7	0.7	0.6	0.6
Precipitation	PrecAvg	1670	13	14	37	88	220	176	307	347	277	128	46	17
	PrecMax	2197	60	94	114	208	605	328	456	662	508	255	215	97
	PrecMin	1348	0	0	0	2	24	31	152	154	97	18	0	0
	PrecStd	225	18	23	33	50	117	74	71	110	96	58	45	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	33.9	36.4	38.3	38.5	35.5	34.6	33.3	33.4	33.0	32.6	31.0
		MCWB	20.0	19.5	21.1	23.1	24.5	25.5	25.4	25.9	25.8	25.0	23.8	22.3
	2%	DB	29.7	32.6	35.1	36.8	35.9	33.7	33.0	32.2	32.5	32.1	31.2	29.7
		MCWB	19.8	19.5	21.1	23.2	24.8	25.4	25.6	25.7	25.5	24.6	23.0	21.4
	5%	DB	28.7	31.4	34.0	35.5	34.3	32.6	31.8	31.2	31.6	31.3	30.2	28.6
		MCWB	19.6	19.6	21.1	23.3	25.0	25.4	25.4	25.6	25.3	24.3	22.6	20.7
	10%	DB	27.6	30.2	32.6	34.2	32.8	31.5	30.7	30.2	30.7	30.3	29.1	27.2
		MCWB	19.3	19.6	21.2	23.5	25.0	25.2	25.2	25.4	25.1	24.1	22.1	20.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	23.4	24.4	26.2	26.7	26.7	26.8	26.7	26.7	26.1	24.8	23.8
		MCDB	27.4	29.9	31.9	33.4	33.4	32.2	31.6	31.3	31.4	31.0	30.1	29.0
	2%	WB	21.4	22.2	23.6	25.3	26.1	26.2	26.2	26.2	26.2	25.5	24.2	22.6
		MCDB	26.8	28.6	30.9	32.7	32.6	31.4	30.8	30.5	30.8	30.3	29.2	27.4
	5%	WB	20.7	21.3	22.9	24.7	25.6	25.8	25.8	25.9	25.8	25.0	23.7	21.8
		MCDB	26.3	27.7	30.2	32.2	31.8	30.7	30.1	30.0	30.2	29.7	28.3	26.4
	10%	WB	20.0	20.5	22.2	24.3	25.2	25.5	25.5	25.6	25.4	24.6	23.1	21.1
		MCDB	25.5	27.2	29.6	31.5	30.9	30.1	29.5	29.3	29.5	28.9	27.2	25.5
Mean Daily Temperature Range	5% DB	MDBR	13.9	15.7	14.8	12.4	9.0	7.1	6.2	6.2	7.2	8.4	10.8	12.1
		MCDBR	15.5	17.5	17.0	14.4	11.5	8.7	8.1	7.7	8.6	9.6	11.9	13.2
		MCWBR	6.1	5.9	5.2	3.9	3.4	2.5	2.5	2.5	2.8	3.1	4.2	5.2
	5% WB	MCDBR	12.5	14.3	13.5	12.1	9.6	7.6	7.1	7.0	7.8	8.4	9.7	11.0
		MCWBR	5.3	5.6	4.9	4.2	3.3	2.6	2.5	2.6	2.8	3.0	3.6	4.4
Clear-Sky Solar Irradiance	taub	0.420	0.502	0.705	0.672	0.491	0.439	0.433	0.456	0.473	0.495	0.418	0.413	
	taud	2.233	2.002	1.557	1.681	2.205	2.384	2.389	2.305	2.232	2.130	2.303	2.277	
	Ebn at Noon	850	795	657	688	817	853	859	845	829	793	846	843	
	Edh at Noon	131	174	281	251	147	122	121	133	141	151	121	122	
All-Sky Solar Radiation	RadAvg	4.46	5.33	5.50	5.95	5.38	4.72	4.27	4.29	4.52	4.40	4.47	4.13	
	RadStd	0.25	0.26	0.31	0.37	0.48	0.33	0.33	0.27	0.20	0.21	0.33	0.29	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.32	+0.78	N/A	N/A	+0.18	+0.14	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.33	+0.63	N/A	N/A	+0.12	+0.19	N/A	N/A	+158	+156

Nomenclature: See separate page